

Subconscious Acquisition of Theoretical Knowledge Through Passive Exposure

A Longitudinal Case Study Proposal

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Research Proposal

Abstract

Research on implicit and subconscious learning has predominantly examined procedural knowledge acquisition, such as the learning of artificial grammars, motor sequences, and perceptual patterns. Very little work has extended these findings into the domain of theoretical knowledge, that is, factual, conceptual, and propositional content of the kind assessed in formal education at GCSE and A-Level. This proposal outlines a longitudinal single-case study designed to investigate whether a student can acquire meaningful theoretical knowledge through daily passive exposure alone, without deliberate study or formal instruction. The participant will listen to 30 minutes of audio recordings about a psychology topic they are not currently studying in school, and read 30 minutes of contextual materials that embed related concepts indirectly, every day for 10 weeks. Recall and understanding will be assessed bi-weekly across four testing points, using a multi-layered assessment that distinguishes recognition from recall and surface knowledge from structural comprehension. The study draws on research in implicit learning, statistical learning, the Input Hypothesis, incidental acquisition, and levels of processing to construct a theoretically grounded design. If the participant demonstrates measurable knowledge gains through passive exposure alone, the findings would extend the literature on subconscious learning into a domain it has rarely addressed directly, and carry practical implications for how educational content might be supplemented through informal channels.

Keywords: *subconscious learning, implicit learning, passive exposure, theoretical knowledge, incidental acquisition, longitudinal case study, psychology education*

1. Introduction

1.1 Background

Most people can recall things they never deliberately set out to learn. A phrase from a song heard repeatedly on the radio, a fact absorbed from a conversation overheard in a queue, a historical detail picked up from a documentary playing in the background. These are not accidents. They are the surface evidence of learning processes that operate below the threshold of deliberate effort. Cognitive psychology has spent decades investigating these processes under labels like implicit learning, statistical learning, and incidental acquisition. The findings are broadly consistent: humans are capable of extracting complex patterns and regularities from their environment without conscious intent and often without awareness that learning has occurred (Reber, 1993; Saffran et al., 1996; Kaufman et al., 2010).

Yet there is a strange limitation in how this research has been applied. The vast majority of studies on implicit and subconscious learning have focused on procedural knowledge, things like motor skills, perceptual discrimination, artificial grammar rules, and sequence learning. These are tasks where the learner acquires know-how through repetition and practice, building up performance through action rather than instruction. What has received far less attention is whether these same subconscious mechanisms can operate on theoretical knowledge, the kind of factual, conceptual, and propositional content that fills textbooks and examination papers in subjects like psychology, history, philosophy, and biology.

This is a meaningful gap. In the UK education system, students at GCSE and A-Level are expected to master large bodies of theoretical content. The dominant mode of learning remains deliberate study: reading textbooks, making notes, answering

practice questions, and revising. Formal instruction assumes that theoretical knowledge requires active, conscious processing to be acquired and retained. But if the subconscious learning mechanisms documented in the cognitive literature are as powerful as the evidence suggests, it is worth asking whether at least some theoretical knowledge might be absorbed through sustained passive exposure, repeated contact with relevant content through reading and listening, without deliberate study, note-taking, or explicit instruction.

1.2 Research Problem

The existing literature presents a paradox. On one hand, implicit learning research shows that humans are remarkably good at extracting structure from their environment without trying (Reber, 1993). Infants can segment speech streams after just two minutes of exposure (Saffran et al., 1996). Adults learn artificial grammar rules without being able to articulate what they have learned (Reber, 1967). On the other hand, educational practice treats theoretical knowledge as something that must be actively constructed through effortful engagement. There is surprisingly little empirical work sitting between these two positions, testing directly whether sustained passive exposure to theoretical content produces measurable knowledge gains. This proposal addresses that gap.

1.3 Research Questions

The study poses three primary research questions:

1. Can a student acquire measurable theoretical knowledge in a subject they have not formally studied, through daily passive exposure to audio and textual materials, without deliberate study or instruction?

2. Does the nature of the knowledge acquired differ across modalities, with direct auditory exposure (listening) producing different patterns of recall and understanding than indirect textual exposure (reading contextual materials)?
3. Does the depth of acquired knowledge increase over time, moving from surface-level recognition toward deeper structural understanding with continued exposure?

1.4 Aims and Objectives

The primary aim is to establish whether subconscious learning processes, well-documented for procedural tasks, also operate meaningfully on declarative theoretical content of the kind assessed in formal education. The objectives are as follows. First, to design and deliver a controlled protocol of daily passive exposure to theoretical content (psychology) over 10 weeks. Second, to measure knowledge acquisition at multiple levels, distinguishing recognition, recall, comprehension, and structural understanding. Third, to track the trajectory of knowledge growth across five bi-weekly assessment points, identifying whether gains appear gradually or show threshold effects. Fourth, to compare the relative contributions of direct auditory exposure and indirect textual exposure to the knowledge acquired. Fifth, to discuss the implications for educational practice and for the broader literature on subconscious learning.

2. Literature Review

2.1 Foundations of Implicit Learning

The modern study of implicit learning begins with Reber's (1967) artificial grammar experiments, in which participants were exposed to strings of letters generated by a finite-state grammar and subsequently asked to classify new strings as grammatical or ungrammatical. The key finding was that participants performed significantly above chance despite being unable to articulate the rules governing the grammar. Reber (1993) later argued that implicit learning is "a fundamental and ubiquitous process" in human cognition, one that operates independently of conscious hypothesis-testing and is phylogenetically older than explicit analytic reasoning. That is a strong claim, and it has been both supported and contested in the decades since.

Support came from an unexpected direction. Saffran et al. (1996) demonstrated that 8-month-old infants could segment a continuous speech stream into word-like units by tracking transitional probabilities between syllables, a capacity they labelled statistical learning. The significance of this finding extends beyond developmental psychology. If infants with minimal cognitive resources can extract structure from continuous input after only two minutes of exposure, it suggests that pattern extraction is a deep property of the human cognitive system, not something that requires mature reasoning or deliberate effort. Perruchet and Pacton (2006) later argued that statistical learning and implicit learning are "two approaches to the same phenomenon," a claim that usefully collapses the terminological boundary while acknowledging that the two research traditions emphasise different mechanisms, the former focusing on computational sensitivity to distributional statistics, the latter on unconscious rule abstraction.

Kaufman et al. (2010) complicated the picture productively by treating implicit learning as an individual-difference variable. They found that implicit learning ability was "largely unrelated to psychometric intelligence" and to working memory, yet was associated with processing speed, verbal analogical reasoning, and foreign-language aptitude. This matters because it suggests that implicit learning is not simply the residue of poor explicit learning or a lower-order process that smarter people outgrow. It appears to be a genuine cognitive capacity, partly independent from the conscious analytic system, yet still predictive of meaningful real-world outcomes. The practical implication is that even high-ability learners may have untapped potential for subconscious acquisition that formal education does not directly engage.

However, not everyone accepts that learning can truly occur without awareness. Schmidt (1990) proposed the Noticing Hypothesis, arguing that "subliminal language learning is impossible" and that for input to become intake, the learner must attend to it at some level. His position does not deny that learning can be unintentional, but it insists that the material must be consciously noticed, even briefly, for it to be encoded. Tomlin and Villa (1994) refined this debate by distinguishing three components of attention: alertness, orientation, and detection. They argued that only detection is necessary for learning, which means that awareness in the full phenomenological sense may be dispensable, but that some form of cognitive registration is not.

This tension is central to the proposed study. Dienes and Berry (1997) addressed it directly by arguing that implicit learning can occur "below the subjective threshold," meaning that learners may acquire knowledge that influences their behaviour without any accompanying subjective experience of knowing. They distinguish this from the objective threshold, below which no information processing occurs at all. The result is a more nuanced picture: learning does not require full conscious awareness, but it may

require some level of perceptual registration. The question for the present study is whether daily passive exposure, where the participant is perceptually engaged with content (listening, reading) but not deliberately trying to learn, falls above or below the threshold at which subconscious acquisition operates.

2.2 Passive Exposure and Knowledge Acquisition: The Input Debate

The strongest theoretical argument for passive exposure as a sufficient learning condition comes from Krashen's Input Hypothesis (1982, 1985). Krashen argued that language acquisition occurs when learners receive "comprehensible input" at a level slightly beyond their current competence ($i+1$), and that this process is largely unconscious. In his view, acquisition through meaningful input is not merely one pathway among many. It is the central mechanism: "we acquire language in one way and only one way: when we understand messages" (Krashen, 1985). Explicit instruction, in Krashen's framework, plays at best a monitoring role, cleaning up output after the fact rather than driving the acquisition process itself.

Krashen's position has been enormously influential and enormously contested. One line of empirical support comes from studies on incidental vocabulary learning, where learners acquire word knowledge as a by-product of reading or listening rather than through direct vocabulary instruction. Elley (1989) found that children who listened to stories read aloud gained vocabulary incidentally, with gains of approximately 15% for target words embedded in the narrative, even though vocabulary was never explicitly taught. Swanborn and de Glopper (1999), in a meta-analysis of 20 studies, confirmed that "incidental word learning does occur during normal reading" and estimated a pickup rate of approximately 15% of unknown words encountered in context. Webb (2007) added that the number of repetitions matters: more encounters

with a word during reading led to greater gains in form recognition, meaning recall, and grammatical knowledge of the word.

The evidence for listening specifically is also positive. Brown et al. (2008) compared three conditions: reading alone, listening alone, and reading-while-listening. All three produced vocabulary gains from incidental exposure, but reading-while-listening was the most effective, suggesting that the combination of modalities creates a stronger encoding opportunity. Vidal (2011) compared reading and listening for incidental vocabulary acquisition in an academic context and found that both produced significant gains, though reading produced marginally higher retention than listening. Her interpretation was that reading allows self-paced processing and re-reading, whereas listening is transient and cannot be revisited. This modality difference is directly relevant to the proposed study, which uses both channels and can therefore observe whether they produce different patterns of acquisition.

However, there are serious objections to treating passive exposure as a sufficient learning mechanism for everything. Nation (2001) cautioned that incidental learning from reading is a "gradual and cumulative" process, noting that "a single encounter with a word in context provides only a small chance of learning." He argued that 10 or more encounters may be needed before a word is reliably known, and that even then the depth of knowledge gained incidentally is usually shallower than what is achieved through deliberate study. Hulstijn (2001) was more direct, arguing that intentional learning is "more effective and efficient" than incidental learning for vocabulary, because deliberate effort focuses attention and creates stronger memory traces. The challenge this poses for the present study is clear: if incidental exposure produces only shallow, fragile knowledge even for vocabulary (a relatively simple domain), can it produce anything at all for more complex theoretical content?

This is where the research leaves a significant gap. Nearly all empirical work on incidental acquisition has focused on vocabulary, a domain where the unit of learning is small, discrete, and repeatable. Theoretical knowledge, by contrast, involves larger conceptual structures: theories, causal relationships, arguments, and interconnected ideas. Whether subconscious mechanisms that work for word-level learning can also support the acquisition of this kind of propositional knowledge is an open question that the literature does not adequately address.

2.3 Memory, Depth, and the Processing Challenge

One of the most influential frameworks for thinking about why some learning sticks and other learning does not is the levels of processing model proposed by Craik and Lockhart (1972). Their central argument was that "the persistence of the memory trace is a positive function of the depth to which the stimulus has been processed." Shallow processing, where the learner engages only with the surface features of the material (its physical form, its sound), produces weak and transient memory traces. Deep processing, where the learner engages with the meaning, implications, and connections of the material, produces durable long-term retention.

This framework presents an immediate theoretical challenge for passive exposure as a learning mechanism. If the participant is listening to audio or reading text without deliberately trying to understand, analyse, or remember, it is reasonable to worry that the processing will be shallow and the resulting knowledge fragile. Craik and Tulving (1975) provided experimental evidence for this concern, showing that incidental encoding under semantic orienting tasks produced better recall than encoding under phonemic or structural tasks. Their results suggest that what matters is not

whether the learner intends to learn, but whether the processing that occurs during exposure engages meaning.

The counter-argument is that passive exposure to meaningful content may naturally produce semantic processing even without deliberate effort. When someone listens to a recording explaining how conformity works, or reads a news article describing bystander apathy, they are processing meaning as part of ordinary comprehension. They do not need to intend to remember the information for semantic engagement to occur. The question is not whether passive exposure can engage deep processing at all, but whether it does so reliably and deeply enough to support lasting recall and understanding.

Memory research also highlights the role of repetition and spacing. Ebbinghaus (1885/1913) established that memory decays rapidly without review, with the sharpest forgetting occurring in the first hours after initial exposure. However, Cepeda et al. (2006) demonstrated in a meta-analysis of 254 observations that distributed practice, spreading learning out over time rather than massing it, produces significantly better long-term retention. The daily exposure protocol proposed in this study implicitly builds in distributed practice: the participant encounters related content every day, and themes recur across recordings and readings. This might partially offset the shallow-processing concern, because even individually weak traces can accumulate and consolidate through repeated contact.

A more challenging objection comes from the testing effect literature. Roediger and Karpicke (2006) demonstrated that actively retrieving information from memory, rather than passively re-reading it, is the most powerful driver of long-term retention. In their study, students who took practice tests retained significantly more material than

students who spent the same time re-studying. This finding directly challenges the logic of passive exposure: if retrieval practice is the engine of durable learning, then simply listening and reading, no matter how often, may be inherently limited. The bi-weekly assessments in the proposed study introduce a retrieval component, but they occur only every two weeks, not daily. Whether the spacing of tests is sufficient to leverage the testing effect alongside passive exposure is an empirical question the study can begin to address.

Tulving's (1972) distinction between episodic and semantic memory is also relevant. Episodic memory records specific events and experiences, while semantic memory stores generalised knowledge that has been abstracted from its original context. The kind of knowledge tested in formal education is largely semantic: you need to know what conformity is and how it works, not where you were when you first heard about it. The question is whether passive exposure produces knowledge that transfers from the episodic context in which it was initially encountered (listening to a recording on Tuesday afternoon) into the decontextualised semantic store that formal assessments tap. If the answer is no, passive exposure might produce the feeling of familiarity without the substance of understanding.

2.4 Theoretical Knowledge and the Declarative Gap

Most of the evidence reviewed so far concerns either procedural knowledge (grammar rules, perceptual patterns) or vocabulary (discrete word-meaning associations). The question of whether subconscious mechanisms can support the acquisition of declarative theoretical knowledge, that is, knowledge of facts, concepts, theories, and their relationships, remains almost entirely unexplored.

Anderson's (1983) ACT-R framework draws a fundamental distinction between declarative knowledge (knowing that) and procedural knowledge (knowing how). Procedural knowledge is acquired through practice and gradually becomes automated. Declarative knowledge, by contrast, is typically acquired through explicit encoding and is stored as propositions or schemas that can be consciously accessed and reported. The implicit learning literature has focused overwhelmingly on the procedural side. Reber (1993) himself acknowledged that implicit learning is best suited to "complex, rule-governed stimulus environments," which describes artificial grammars and statistical sequences but does not obviously describe the kind of propositional content that fills a psychology syllabus.

This is a genuine limitation of the current evidence base. Eraut (2000) highlighted that professional practice involves a mixture of formal and informal knowledge, and that much of the informal knowledge professionals rely on is acquired "implicitly" through participation in everyday work. But Eraut's account, while persuasive as a description of workplace learning, does not offer a controlled test of whether factual academic content can be acquired the same way. The closest empirical analogue is the vocabulary acquisition literature reviewed above, which shows that individual pieces of knowledge can be picked up incidentally but that the process is slow, cumulative, and often shallow.

Smalle et al. (2022) contribute an important nuance. They demonstrated that adults retain robust implicit statistical learning abilities, but that these abilities can be "altered" or suppressed by cognitive control. In their study, adults who performed a concurrent task that depleted executive resources actually showed better statistical learning than adults who were free to engage their full cognitive apparatus. The implication is that trying too hard, engaging excessive conscious monitoring, may

interfere with the subconscious pattern extraction that drives implicit learning. This is a provocative finding for education, because it suggests that the deliberate, effortful study strategies that formal education promotes might actually suppress the very mechanisms that could support learning without effort.

This study proposes to address the declarative gap directly. By exposing a participant to theoretical content through passive channels (listening and reading) and testing whether factual and conceptual knowledge accumulates over time, the study can provide the first empirical evidence, however preliminary, on whether subconscious acquisition extends beyond procedural and lexical domains into the kind of theoretical content that formal education treats as requiring conscious effort.

2.5 Modality and Dual-Channel Processing

The proposed study uses two modalities of exposure: direct auditory input (recordings about the topic) and indirect textual input (reading materials that embed related concepts). This design raises the question of how modality affects subconscious knowledge acquisition.

Paivio's (1986) dual coding theory argues that information is encoded through two partially independent systems: a verbal system for language-based information and an imagery system for non-verbal perceptual information. When both systems are engaged, the resulting memory trace is stronger and more retrievable. While the study does not use visual imagery in the traditional sense, the combination of auditory and textual exposure creates two encoding pathways: one through speech comprehension and another through reading comprehension. These are neurologically and cognitively distinct processes, even though both handle verbal information.

Conway and Christiansen (2005) tested statistical learning across modalities, visual, auditory, and tactile, and found that "statistical learning may be modality-constrained." Their participants showed different patterns of learning depending on the modality of presentation, and learning in one modality did not transfer fully to another. This suggests that the auditory and textual components of the proposed study may contribute different kinds of knowledge, or at least encode the same knowledge in different ways. It also raises the possibility that the combination of modalities could produce a richer and more durable representation than either alone.

This connects to Brown et al.'s (2008) finding that reading-while-listening outperformed both reading alone and listening alone for vocabulary acquisition. If this modality advantage extends to theoretical content, the dual-channel design may amplify the learning effect, assuming an effect exists at all. However, it is worth noting a critical difference. In Brown et al.'s study, participants read and listened to the same content simultaneously. In the proposed study, the audio content directly addresses the topic while the reading materials embed concepts indirectly. This asymmetry is deliberate: it allows the study to compare direct passive exposure (listening) with indirect passive exposure (reading contextual materials that hint at the topic without explicitly teaching it). Whether this comparison strengthens or complicates the design is something the pilot phase will help determine.

2.6 Summary and Research Gap

The literature reviewed above establishes several things. First, humans possess robust subconscious learning mechanisms that extract patterns and structure from the environment without deliberate effort (Reber, 1967, 1993; Saffran et al., 1996; Kaufman et al., 2010). Second, passive exposure through listening and reading can

produce incidental knowledge gains, particularly for vocabulary (Elley, 1989; Swanborn & de Glopper, 1999; Webb, 2007; Brown et al., 2008; Vidal, 2011). Third, the durability and depth of incidentally acquired knowledge are contested, with critics arguing that shallow processing (Craik & Lockhart, 1972), infrequent retrieval (Roediger & Karpicke, 2006), and the superiority of deliberate effort (Hulstijn, 2001) limit the practical value of passive exposure. Fourth, the role of awareness remains unresolved, with Schmidt (1990) insisting on noticing as a necessary condition and Dienes and Berry (1997) arguing that learning can occur below the subjective threshold. Fifth, and most critically, almost all empirical work has focused on procedural or lexical knowledge. The question of whether subconscious mechanisms can support the acquisition of declarative theoretical knowledge, the kind assessed in formal education, has not been adequately tested.

This study addresses that gap directly. It asks whether daily passive exposure to theoretical content, delivered through auditory and textual channels, produces measurable knowledge gains in a participant who has received no formal instruction in the subject.

3. Methodology

3.1 Research Design

The study uses a longitudinal single-case design with repeated measures. A single participant will receive daily passive exposure to theoretical content for 10 weeks, with assessments administered at five points: baseline (Week 0, before exposure begins), Week 2, Week 4, Week 6, Week 8, and Week 10 (final). This design allows the tracking of knowledge acquisition trajectories over time and the identification of growth patterns, plateaus, or threshold effects.

The single-case design is appropriate for this exploratory study for several reasons. First, the phenomenon under investigation (subconscious acquisition of theoretical knowledge through passive exposure) has not been empirically tested before, and a detailed longitudinal observation of one participant's experience will generate richer and more interpretable data than a shallow between-groups comparison at this stage. Second, the daily commitment required, one hour per day for 10 weeks, makes large-scale recruitment impractical for a student-led project. Third, the case study tradition in educational and psychological research is well-established (Yin, 2018; Stake, 1995) and is suited to investigating complex processes in naturalistic conditions.

3.2 Participant Selection

One participant will be recruited through convenience sampling. The selection criteria are as follows:

1. The participant must be of secondary school age (Years 10–13, aged 14–18), currently enrolled in a UK school or sixth form.

2. The participant must NOT be currently studying Psychology, nor have previously studied Psychology at GCSE, A-Level, or equivalent.
3. The participant must not be receiving private tutoring in Psychology or related subjects.
4. The participant must have no diagnosed learning difficulties that would significantly affect reading comprehension or auditory processing.
5. The participant must be able to commit to one hour of daily engagement for 10 consecutive weeks.

These criteria ensure that any knowledge gained during the study cannot be attributed to formal instruction or prior learning. A pre-study screening questionnaire will confirm eligibility and establish the baseline level of existing knowledge about psychology, which is expected to be near zero for the specific content tested.

3.3 Topic Selection and Justification

The topic for the study is introductory psychology, specifically five areas drawn from the AQA A-Level Psychology specification: memory models, social influence, attachment theory, approaches in psychology, and psychopathology. This topic was selected for the following reasons:

1. It is purely theoretical and knowledge-based. Answering questions about Milgram's obedience study or Bowlby's attachment theory requires knowing facts and understanding concepts, not solving mathematical problems or performing practical procedures.

2. It does not require "active thinking" in the sense of calculation or algorithmic reasoning. A student cannot work out the details of the multi-store model of memory from first principles; they must have encountered the content.
3. The content is standardised. The AQA specification provides a clear and assessable body of knowledge, which allows the construction of valid and reliable assessments.
4. Materials are readily available. There are high-quality audio and textual resources for psychology that can be adapted for the study.
5. The content is varied enough to sustain 10 weeks of daily exposure without excessive repetition of identical material.

Each two-week block will focus on one of the five topic areas, allowing cumulative tests to assess both current-block learning and retention of previous blocks.

3.4 Materials Development

3.4.1 Audio Recordings (Direct Exposure)

A set of audio recordings will be produced for each two-week block, totalling approximately 10 hours of content per block (30 minutes per day x 14 days). The recordings will be narrated summaries of key theories, researchers, studies, and findings from the relevant area of psychology. The tone will be conversational and engaging, similar to a documentary or podcast, rather than a formal lecture. Recordings will be produced in British English.

Crucially, the participant will be instructed to listen without taking notes, without pausing to look things up, and without deliberately trying to memorise. They will be told to listen as they would to a podcast or radio programme. The recordings

will be varied, not simply repeating the same script each day, but covering the same conceptual territory from different angles, examples, and narrative structures. This ensures repeated exposure to core concepts without monotonous verbatim repetition.

3.4.2 Reading Materials (Indirect Exposure)

The reading materials are designed to differ from the audio in a critical way. Where the audio directly explains psychological theories and findings, the reading materials will embed psychological concepts within real-world contexts without explicitly naming or teaching them. This is what is meant by materials that "hint at" the topic.

For example, during the social influence block, the participant might read:

- A news article about jury deliberation and group pressure (embedding conformity without naming it)
- A short story about a person who follows a harmful order at work (embedding obedience without referencing Milgram)
- A magazine piece about how social media creates echo chambers (embedding informational social influence)
- A case study of a corporate scandal where nobody spoke up (embedding groupthink and bystander effects)

The participant will be told that the reading component is part of a "general media literacy" exercise. They will read normally, without highlighting, note-taking, or deliberate study. This indirect exposure channel tests whether theoretical concepts can

be absorbed even when they are not explicitly labelled or taught, mirroring the way people naturally encounter ideas through general reading.

Each day's reading will be curated and pre-selected. All materials will be screened for readability (targeting a Flesch-Kincaid grade level of 8–10, appropriate for the age range) and for accurate implicit representation of the psychological concepts.

3.5 Daily Protocol

The participant will complete one hour of exposure per day, seven days a week, for 10 consecutive weeks:

Component	Duration	Format	Instruction
Audio listening	30 minutes	Headphones, quiet environment	Listen naturally, no notes, no pausing
Reading	30 minutes	Printed or screen-based	Read normally, no highlighting or note-taking

The order of components (audio first or reading first) will be consistent for the participant across all days to reduce order-effect confounds. A brief daily compliance log will be completed by the participant, recording the start time, end time, and any disruptions. This log will not require reflection on content, only confirmation that the session was completed.

To minimise external contamination, the participant will be asked to avoid deliberately seeking out information about psychology during the study period. A weekly check-in (5 minutes) will include a brief interview about any incidental exposure that occurred outside the study (a conversation about psychology, a relevant TV programme, etc.). These will be logged as potential confounds.

3.6 Testing Procedure

Assessments will be administered at baseline (Week 0) and then bi-weekly at Weeks 2, 4, 6, 8, and 10. Each assessment lasts approximately 45–60 minutes and consists of four parts:

Part	Format	What It Measures	No. of Items
A: Recognition	Multiple choice (4 options)	Surface-level recognition of terms, names, concepts	20 items
B: Cued Recall	Short-answer questions	Ability to retrieve specific facts and details	10 items
C: Comprehension	Extended-response questions	Understanding of theories, ability to explain and apply	3 questions
D: Concept Map	Open concept mapping	Structural understanding, how concepts relate	1 exercise

Part A tests the shallowest level of knowledge: can the participant recognise a term or idea when it is presented with distractors? This is the level most likely to be supported by passive exposure. Part B tests recall, which is harder because it requires the participant to generate an answer rather than simply recognise one. Part C tests understanding and application, requiring the participant to explain a theory or apply a concept to a novel scenario. Part D, the concept map, tests whether the participant has developed a structural understanding of how ideas relate, which would indicate deeper learning beyond isolated facts.

Each bi-weekly assessment will contain items from all topic blocks covered so far. This means that the Week 4 test will include items from Blocks 1 and 2, the Week 6 test will include items from Blocks 1–3, and so on. This cumulative testing allows the observation of both current learning and long-term retention.

Items will be written by the researcher and reviewed by a psychology teacher or tutor familiar with the AQA specification, to ensure content validity. A small pilot test with 2–3 individuals who have A-Level psychology knowledge will be used to calibrate item difficulty.

3.7 Data Collection

Data will be collected from five sources:

1. Assessment scores (Parts A–D, at each testing point)
2. Daily compliance logs (time, duration, disruptions)
3. Weekly check-in interviews (5 minutes, recorded, inquiring about incidental exposure)
4. Concept maps (photographed or digitally submitted)
5. A post-study semi-structured interview (30 minutes, exploring the participant's subjective experience of the learning process)

The post-study interview will explore whether the participant was aware of learning, which topics they felt more or less familiar with, and whether they noticed any changes in their thinking or knowledge during the study. This qualitative component is important because it addresses the subjective dimension of subconscious learning: did the participant feel like they were learning, or did the knowledge appear without any accompanying sense of acquisition?

3.8 Data Analysis

Quantitative analysis will be primarily descriptive, given the single-case design. The main analyses will be:

1. Mean scores for each assessment part (A–D) at each time point, plotted to show learning trajectories.
2. Percentage change from baseline to final assessment, calculated for each part separately.
3. Within-part comparisons: Do recognition scores (Part A) grow faster than recall scores (Part B) or comprehension scores (Part C)?
4. Retention analysis: Do scores on earlier blocks decline, hold steady, or continue to improve across later assessments?
5. Concept map complexity: Number of nodes, links, and cross-links at each time point, scored using Novak and Gowin's (1984) hierarchical scoring method.

While inferential statistics are not appropriate for a single-case design, visual inspection of trajectories, reliable change indices, and comparison against the baseline score will be used to assess whether observed changes are meaningful. A tau-U statistic may also be computed if the data series is long enough, providing a non-parametric estimate of trend.

Qualitative data from the weekly check-ins and post-study interview will be analysed using thematic analysis (Braun & Clarke, 2006), focusing on themes related to awareness of learning, perceived difficulty, and subjective sense of knowledge growth.

3.9 Validity and Reliability

Several measures will be taken to strengthen validity:

Internal validity: The main threat is that the participant may encounter psychology content outside the study. Weekly check-ins and a content diary will monitor for this, and any external exposure will be recorded and discussed in the analysis as a potential confound.

Content validity: Assessment items will be aligned with the AQA A-Level Psychology specification and reviewed by a subject specialist.

Construct validity: The four-part assessment structure distinguishes recognition, recall, comprehension, and structural understanding, allowing analysis of whether passive exposure supports different levels of knowledge to different degrees.

Reliability: Assessments at different time points will use parallel forms (different items covering the same content) to avoid practice effects from seeing the same questions repeatedly.

The main limitation is external validity. Findings from a single case cannot be generalised to other individuals, topics, or age groups. The study is explicitly exploratory, designed to establish whether the phenomenon exists at all, not to quantify its magnitude across populations.

3.10 Ethical Considerations

The study involves mild incomplete disclosure. The participant will be told that the study investigates "how people engage with different types of media over time" rather than that it is studying subconscious learning. This is necessary because telling

the participant the true purpose would introduce demand characteristics: knowing that the study aims to test whether they learn without trying might cause them to either deliberately try to learn or deliberately resist learning. Either would compromise the validity of the findings.

This approach is consistent with the British Psychological Society's Code of Ethics and Conduct (BPS, 2021), which permits incomplete disclosure where: (a) the research could not be conducted otherwise, (b) the deception is not likely to cause distress or harm, and (c) a full debrief is provided afterwards. All three conditions are met.

The participant (and parent/guardian, if the participant is under 16) will provide written informed consent, will be told they can withdraw at any time without penalty, and will be given a full debrief at the end of the study explaining the true purpose, why incomplete disclosure was necessary, and what the study found. During the debrief, the participant will be given the opportunity to ask questions and to withdraw their data if they object to the true purpose. All data will be anonymised and stored securely in compliance with the UK Data Protection Act 2018 and GDPR. No data will be shared with third parties.

4. Timeline

Phase	Duration	Activities
Preparation	Weeks -6 to -1	Ethics approval, participant recruitment, materials development, pilot testing of assessments
Baseline	Week 0	Pre-study screening, baseline assessment (all four parts), orientation session
Block 1: Memory	Weeks 1–2	Daily exposure (audio + reading), compliance logging, Assessment 1 at end of Week 2
Block 2: Social Influence	Weeks 3–4	Daily exposure, weekly check-in, Assessment 2 at end of Week 4
Block 3: Attachment	Weeks 5–6	Daily exposure, weekly check-in, Assessment 3 at end of Week 6
Block 4: Approaches	Weeks 7–8	Daily exposure, weekly check-in, Assessment 4 at end of Week 8
Block 5: Psychopathology	Weeks 9–10	Daily exposure, weekly check-in, Assessment 5 at end of Week 10
Post-study	Weeks 11–12	Post-study interview, full debrief, data analysis and write-up

Total duration from preparation to write-up is approximately 18 weeks.

5. Expected Outcomes

Based on the literature, several outcomes are anticipated:

First, the participant is expected to show measurable gains in recognition (Part A) from the earliest assessments. Recognition memory is the most sensitive to passive exposure, and even single encounters with information can produce above-chance recognition (Swanborn & de Gloor, 1999).

Second, recall gains (Part B) are expected to appear later and to be smaller than recognition gains. Recall requires deeper encoding and stronger memory traces, which may develop more slowly under passive exposure conditions.

Third, comprehension and application scores (Part C) are the most uncertain. If passive exposure supports only surface-level encoding, these scores may remain near baseline throughout. If passive exposure supports deeper processing, as Craik and Lockhart (1972) would predict for semantically rich content, modest gains may appear in later assessments.

Fourth, concept map complexity (Part D) is expected to increase gradually if the participant develops structural understanding, but this is the least likely outcome under passive conditions.

Fifth, the post-study interview is expected to reveal that the participant has some awareness of having absorbed information, but may be unable to identify when or how specific pieces of knowledge were acquired, consistent with the phenomenology of implicit learning described by Dienes and Berry (1997).

If the study produces null results, with no meaningful knowledge gains observed, this would also be a valuable contribution. It would suggest that subconscious learning mechanisms, however powerful for procedural tasks, do not extend to declarative theoretical content, at least not under the conditions tested. This would support the position that theoretical knowledge genuinely requires deliberate effort, and that the analogy between procedural implicit learning and academic knowledge acquisition does not hold.

6. Limitations

Several limitations should be acknowledged. The single-case design means that findings cannot be generalised beyond the individual participant. Different individuals may respond differently to passive exposure depending on their prior knowledge,

cognitive profile, learning style, and engagement. A future study with multiple participants or a randomised controlled design would be needed to test generalisability.

The 10-week duration, while sufficient for a preliminary investigation, is short relative to the timescales over which academic learning normally occurs. A-Level students study psychology over two years. Whether 10 weeks of passive exposure can meaningfully approximate the depth of learning achieved through formal study over a much longer period is doubtful, and the study does not claim otherwise.

The mild deception creates an ethical and methodological tension. If the participant is not told the true purpose, their engagement with the materials may be more "natural." But if they suspect the true purpose, or if they simply enjoy the content and begin to engage with it more actively, the boundary between passive and active exposure becomes blurred. The weekly check-ins will attempt to monitor for this, but self-reports of engagement may be unreliable.

The division between "direct" audio and "indirect" reading materials is an interesting design feature, but it also introduces a confound. Any differences in learning attributable to modality could also reflect differences in content directness. A future study could cross the design, with both direct and indirect materials in both modalities, to isolate the modality effect from the directness effect.

Finally, the assessments themselves may promote learning. The testing effect (Roediger & Karpicke, 2006) means that bi-weekly assessments are not purely passive measurements. They actively engage retrieval, and this retrieval practice may contribute to knowledge consolidation. This is unavoidable in a study that needs to measure learning at multiple time points, but it means that any observed gains cannot

be attributed solely to passive exposure. They reflect the combined effect of exposure and periodic testing.

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